

Thank you for purchasing a FANUC Robot.

READ THIS FIRST

Use this guide to unpack, install, turn on, and jog your new FANUC robot.

The appropriate level of safety for your application and installation can best be determined by safety system professionals. A qualified electrician should apply power. FANUC recommends that each customer consult with such professionals in order to provide a safe application, and take the necessary FANUC training course to operate the robot safely. Refer to your FANUC documentation provided with your robot for safety procedures.

Required Tools

- ◆ Straight-head screwdriver ◆ Torque wrench
- ◆ Cross-head screwdriver ◆ Metric hex-head wrenches

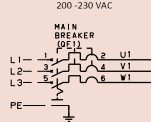
Required Items

- ◆ Properly rated transport equipment as specified
- ◆ Customer-supplied properly rated power source as specified on the Controller Power Source Label
- ◆ Customer-supplied properly rated input power cable; conductor size of AWG10; terminal size is M5
- ◆ Multimeter ◆ Rigid platform to hold robot
- ◆ 4-M16 x 65mm bolts ◆ Loctite #243
- ◆ 4-M16 high strength washers

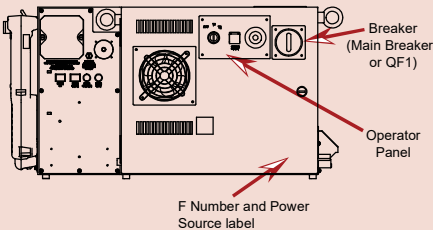
Optional Items

- ◆ Strain relief cable grip
- ◆ Robot locator pins

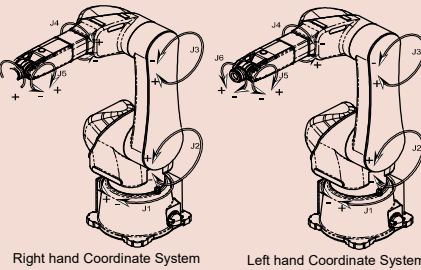
Three Phase



Controller



Robot (mechanical unit) with joints indicated

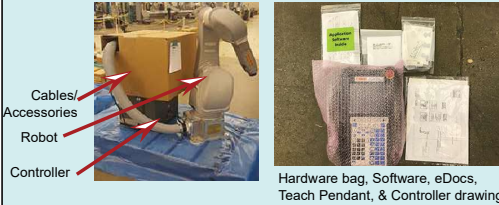


P-40iA 6-Axis Robot

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A. Unpack/Install your Robot

Shipment Contents



Hardware bag, Software, eDocs, Teach Pendant, & Controller drawing
Keep this information in a safe, known location.

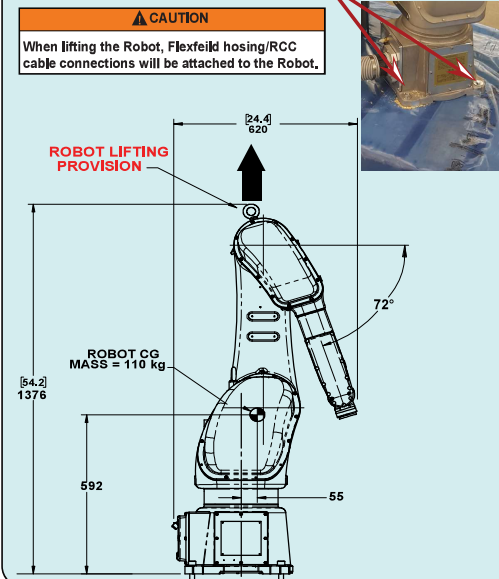
WARNING

Before moving heavy equipment, check and tighten any loose bolts. Do not pull eyebolts sideways. Otherwise, you will injure personnel or damage equipment.

1. Open all of the boxes included in your shipment.
2. Remove the plastic covering from the robot.
3. Use properly rated equipment to move the robot to its new location.
4. Secure Robot from "Robot lifting provision" eye bolt. Make sure lifting strap is secure and taut while lifting.
5. Remove the bolts holding the robot to the transport skid. Then lift and place into the new location.

CAUTION

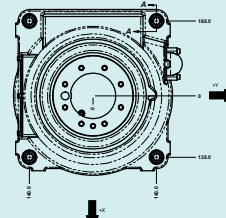
When lifting the Robot, Flexfield hosing/RCC cable connections will be attached to the Robot.



6. Use (4) M16 bolts with washers to bolt the robot to a rigid platform or table that will prevent any motion of the robot base. See diagram for P-40iA mounting patterns.

- Refer to the robot model-specific Operator's Manual for force and moment data, and mounting pattern information.

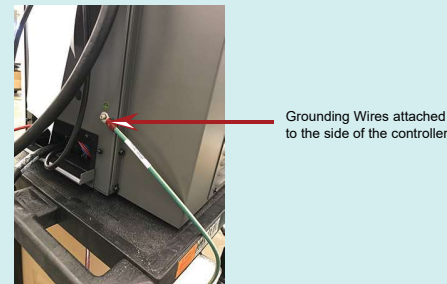
NOTE: Be careful to avoid interference with bolts or connections.



7. Secure the robot using M16 bolts.

280 N-m (206 ft-lbs torque)

8. Connect the ground wires together.



9. Insert the blue tubing into the gray fitting (input air pressure: 65-75 PSI). Completed connections are shown below.

Robot Purge Air Supply:

Minimum Flow switch Activation: 566 SLPM
Maximum Flow switch Activation: 795 SLPM
Air Supply Pressure: 65 - 75 PSI Dynamic
Maximum Air Supply Temp: 45°C
Maximum Maintenance Flow: 70 - 450 SLPM



Humidity: -40°C Pressure Dew point at 621 kPa
Maintenance Pressure: 8 - 12in of H2O
Minimum Overpressure: 0.69in of H2O
Maximum Overpressure: 2.0 PSI
Maximum Leakage flow: 200 SLPM
Operating Ambient Temp: 0° - 45°C
Filtration: 5 Micron Absolute
Exhaust Pressure: 1.75in of H2O

Robot installation is complete. You are now ready to go to B. Unpack/Install your Controller.

B. Unpack/Install your Controller

1. Remove the bolts holding the controller to the skid.



2. Use properly rated equipment to remove the controller from the skid using eyebolts.

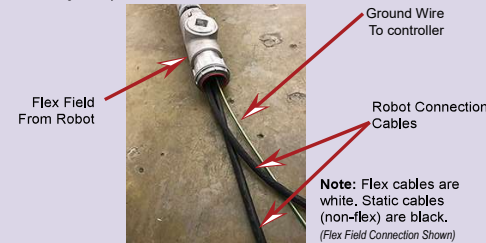
WARNING

Before moving heavy equipment, check and tighten any loose bolts. Do not pull eyebolts sideways. Otherwise, you will injure personnel or damage equipment.

Crane Capacity: >150kg
Sling Capacity: >150kg

3. Place the controller in a safe location where the grounding wire and RCC can reach the robot, and the controller is safely reachable from outside of the robot work area.

Note: Cables are customer-specific. Your cable installation might vary from what is shown here.



Note: Flex cables are white, Static cables (non-flex) are black. (Flex Field Connection Shown)

4. Follow steps 5a - 5i to provide power to the controller.

WARNING

Lethal voltage is present in the controller whenever it is connected to a power source. Be extremely careful to avoid electrical shock. Lock out and tag out the power source at the controller according to the policies of your plant.

- a. Find the power source needed for your controller on the door near the F Number. See the diagram to the left for location. Connect the controller to power.
- b. Open the controller door by using a straight-head screwdriver to turn the latches counter-clockwise, then turn the Breaker counter-clockwise past its off position.
- c. Route the power input cable through the rear of the controller by removing the white plugs.
- d. To connect power, remove the plastic cover from above the Breaker (see the diagram below.) To remove the cover, insert a straight-head screwdriver into the opening, and push the tab to the left allowing the cover to slide forward.



Breaker with cover removed



Go to Page 2

- e. Connect power to the Breaker with the user-supplied input power cable. For three phase, connect L1 to 1, L2 to 3, and L3 to 5. Connect the ground to the plate located to the right of the Breaker. (Reference 3-phase diagram on far left of page 1 of this guide)



Breaker connection with customer-supplied input power



Breaker with cover reinserted

- f. After the connections are made, reinsert the plastic cover.

- g. Use a multimeter to verify the correct voltage is present.

- h. Before you close the controller door, retrieve the set of keys from inside the door, and store them in a known location outside of the controller for use later in this guide.



Key location inside controller.

- i. Close the controller door and use a straight-head screwdriver to turn the Latch on the outside of the controller counter-clockwise.

5. Connect the teach pendant to its cable.



Controller installation is complete. DO NOT TURN ON the CONTROLLER UNTIL YOU are INSTRUCTED TO DO SO. Go to **C. Turn on your Controller**.

C. Turn on your Controller

1. Turn on the controller by rotating the Breaker clockwise to ON.

WARNING

Lethal voltage is present in the controller whenever it is connected to a power source. Be extremely careful to avoid electrical shock. Lock out and tag out the power source at the controller according to the policies of your plant.

2. Follow Purge Sequence (steps 13a-i). The button/lights are shown below.



Purge Sequence

- The operator connects the air supply to the robot. The pressure switch closes and remains closed as long as air is supplied.
- The operator provides power to the controller terminals. After power is provided, the Purge Fault indicator turns on.
- The operator presses and holds the Purge Enable Button.
- The Purge Control Module switches the solenoid valve to distribution lines to start the flow of purge air. Air will be distributed throughout the robot cavity by distribution lines.
- Flow is monitored by the flow switch at the air outlet. When the flow reaches 560 L/min, the flow switch closes the electronic contacts. Flow Indication will come on and the control module will start timing. At the same time, Purge Fault indication turns off.
- When Flow Indication comes on, the operator can release the Purge Enable Button.
- Purge flow continues for 1.7 minutes to ensure air changes. Time is measured by hardware logic in the purge control module.
- After 1.7 minutes, purge is complete. The purge control module switches the solenoid valve to maintenance air, closes the output contacts to provide power to the robot controller, and turns the Purge Complete Indication on. The air pressure in the cavity decreases, and the check valve closes to prevent loss of maintenance air.
- The robot can be turned on. As long as purge is complete, the robot may be turned on and off as desired.

Purge is complete. You are now ready to go to **D. Jog your Robot**.

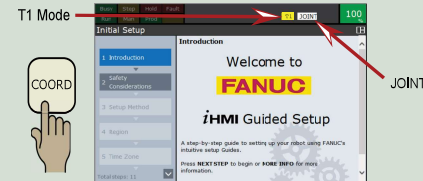
D. Jog your Robot

1. Turn the Breaker On. Allow the Controller to power up. Locate the Mode switch on the Controller Operator Panel. Use the key retrieved in Step 4h to switch to T1 mode. T1 mode limits the robot speed to 250mm/sec and requires robot control to be at the teach pendant only.



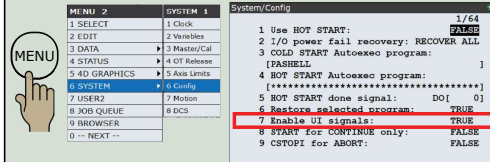
2. Pick up the teach pendant and look at the screen. Verify that the controller is in T1 Mode. If it is not, repeat step 1.

3. Press the COORD key on the teach pendant until JOINT is displayed.

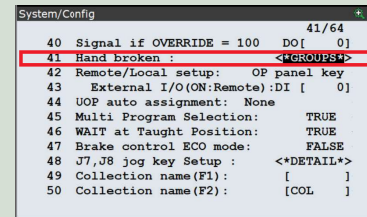


4. To jog the robot in JOINT mode:

- Turn the teach pendant ON/OFF switch to the ON position, press MENU, move the cursor to ALARM, and press ENTER.
- To toggle between alarm history and active alarm display, press F3.
- If the error MCTL-013 ENBL Input is off is displayed in the alarm history when you try to move the robot (it will not display as an active alarm) you must disable UI. To do this,
 - Press MENU, select 0 (Next), and select SYSTEM (6), and then select CONFIG (6) as shown to the right.
 - Use the teach pendant arrow keys to move the cursor to the Enable UI signals option.
 - Use the teach pendant keys to set the option to FALSE.



- If the error SRVO-102 Hand broken is displayed, disable it by moving the cursor to the Hand broken option in the System/Config screen, pressing ENTER, and then disabling it.



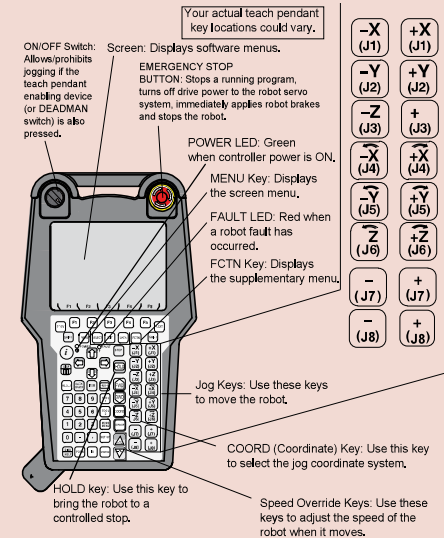
- Ensure the Emergency Stop buttons on the teach pendant and controller are released (popped out).

- Press and hold a DEADMAN switch on the back of the teach pendant in its middle position. Press RESET on the teach pendant to clear alarms.

- To move each joint axis of the robot (jog the robot in JOINT mode), press and hold SHIFT and press each jog key one at a time and watch the robot joint move. Refer to the Robot (Mechanical Unit) with Joints indicated diagram on Page 1 of this guide.

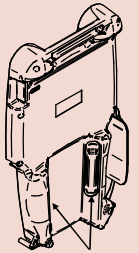
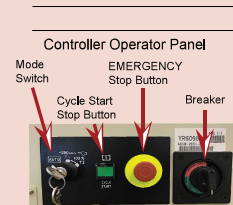
- Ensure you are able to jog each robot joint.

Turning on and Jogging the robot is complete. You are now ready to perform software installation, if necessary. Refer to the **Controller-specific Software Installation** manual.



Refer to your Application-Specific Setup and Operations Manual for more information on each teach pendant key.

iPendant (Teach Pendant)



For more information

To access controller, robot, or application-specific manuals, scan the QR Code to the right or go to <https://www.fanucamerica.com/>.



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